Study Guide: Microbes in Human Welfare

This study guide provides a comprehensive review of the roles microorganisms play in enhancing human welfare across household products, industrial applications, sewage treatment, energy production, and agriculture.

Microbes in Human Welfare: Comprehensive Class 12 Biology Notes

1. Microbes in Household Products

Microbes play a daily role in preparing many of our common foods through the process of fermentation.

- **Curd:** A small amount of curd added to milk acts as an inoculum containing **Lactic Acid Bacteria (LAB)**, such as *Lactobacillus*. LAB produces lactic acid, which coagulates and partially digests milk proteins (casein). It enhances the nutritional quality of the curd by increasing **Vitamin B12** and helps inhibit disease-causing microbes in the human gut.
- **Dough:** *Saccharomyces cerevisiae* (baker's yeast) ferments dough for making bread, idli, and dosa. The characteristic puffed-up appearance of the dough is due to the release of **carbon dioxide (CO_2)** gas during fermentation.
- **Cheese:** Different microbes impart specific textures and flavors to cheese.
    - **Swiss Cheese:** Known for its large holes, which are caused by massive CO_2 production by the bacterium *Propionibacterium shermanii*.
    - **Roquefort Cheese:** Ripened by a specific fungi, *Penicillium roqueforti*.
- **Toddy:** A traditional South Indian drink created by fermenting the sap of palm trees.

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2. Microbes in Industrial Products

Microbes are grown in large vessels called fermentors to synthesize products on a commercial scale.

A. Fermented Beverages

- *Saccharomyces cerevisiae* (brewer's yeast) is used to ferment fruit juices and malted cereals to produce ethanol.
- **Without Distillation:** Wine and beer.
- **With Distillation:** Whiskey, brandy, and rum (distilling the fermented broth increases the alcohol concentration).

B. Antibiotics

- **Penicillin** was the first antibiotic discovered by Alexander Fleming from the mould *Penicillium notatum*.
- Its full potential was established by Ernest Chain and Howard Florey, and the three scientists shared the Nobel Prize in 1945. It was famously used as a "Wonder Drug" to treat wounded American soldiers during World War II.

C. Chemicals (Acids) Different microbes are used to commercially produce organic acids:

- *Aspergillus niger* (Fungus) $\rightarrow$ **Citric acid**
- *Acetobacter aceti* (Bacterium) $\rightarrow$ **Acetic acid**
- *Clostridium butylicum* (Bacterium) $\rightarrow$ **Butyric acid**
- *Lactobacillus* (Bacterium) $\rightarrow$ **Lactic acid**

D. Enzymes & Bioactive Molecules

- **Lipase:** Used in detergent formulations to remove oily and greasy stains from laundry.
- **Pectinase and Protease:** Used to clarify bottled fruit juices, making them clearer than homemade juices.
- **Streptokinase:** Produced by the bacterium *Streptococcus*. It is modified by genetic engineering and used as a **"clot buster"** to remove blood clots from the blood vessels of patients suffering from myocardial infarction (heart attack).
- **Cyclosporin A:** Produced by the fungus *Trichoderma polysporum*. It acts as an **immunosuppressive agent** for patients undergoing organ transplantation.
- **Statins:** Produced by the yeast *Monascus purpureus*. Used as a **blood cholesterol-lowering agent** because it competitively inhibits the enzyme responsible for cholesterol synthesis.

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3. Microbes in Sewage Treatment

Sewage contains large amounts of organic matter and pathogenic microbes. It is treated in Sewage Treatment Plants (STPs) to lower its **Biochemical Oxygen Demand (BOD)**—a measure of organic matter present in the water. Higher BOD means higher polluting potential.

Flowchart of Sewage Treatment:

1. Primary Treatment (Physical Process) $\downarrow$ *Sequential Filtration:* Removes floating debris (like garbage) $\downarrow$ *Sedimentation:* Removes heavy solid waste, grit, and soil $\downarrow$ *Result:* Heavy settled solids form the **Primary Sludge**, while the floating liquid becomes the **Primary Effluent**.

2. Secondary Treatment (Biological Process) $\downarrow$ *Aeration:* Primary effluent is pumped into large aeration tanks with constant mechanical agitation and air supply. $\downarrow$ *Floc Formation:* Vigorous growth of aerobic microbes forms **"Flocs"** (masses of bacteria bound by fungal filaments into a mesh-like structure). $\downarrow$ *BOD Reduction:*

These flocs consume the major organic matter, significantly reducing the wastewater's BOD. $\downarrow$ *Sedimentation:* Once BOD is reduced, the effluent goes to a settling tank. Flocs settle down to form **Activated Sludge**. $\downarrow$ *Anaerobic Digestion:* A small part of activated sludge acts as an inoculum for the aeration tank. The rest is pumped into large **Anaerobic Sludge Digesters**. Here, anaerobic bacteria digest the aerobic microbes, producing a mixture of gases (**Biogas**) like Methane, CO_2, and H_2S.

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4. Microbes in Biogas Production

- Biogas is a highly inflammable, clean, and renewable energy source consisting primarily of **Methane (60-70%)**, Carbon dioxide (30-40%), and traces of Hydrogen and Hydrogen Sulphide.
- **Methanogens** (e.g., *Methanobacterium*) are anaerobic bacteria that breakdown cellulosic organic matter to produce methane. They are naturally found in anaerobic sewage sludge and the rumen (stomach) of cattle.

Diagrammatic Layout of a Biogas Plant:

- **Dung + Water Chamber (Mixing Tank):** Cattle excreta (gobar) and water are mixed into a slurry and fed into the plant.
- **Digester:** A deep concrete tank (10-15 feet) where methanogens break down the slurry in the absence of oxygen.
- **Floating Gas Holder:** A cover that rises as biogas accumulates inside the digester.
- **Outlet Pipe:** Channels the produced biogas to nearby households for heating and cooking.
- **Sludge Chamber:** Collects the leftover, decomposed slurry, which is removed and used as an excellent biofertilizer (manure). *(Note: Biogas technology in India was pioneered by IARI and KVIC.)*

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5. Microbes as Biocontrol Agents

Biocontrol is an ecological approach to managing pests and plant diseases using natural predators instead of toxic, polluting chemical pesticides.

- **Ladybirds and Dragonflies:** Naturally control populations of aphids and mosquitoes.
- **Bacillus thuringiensis (Bt):** A bacterium whose dried spores are sprayed on crops (like brassicas) to combat caterpillar infestations. When eaten, the bacterial toxin kills the larvae by destroying their midgut, without harming other beneficial insects. Genetic engineering has allowed the *Bt* gene to be directly inserted into crops (e.g., Bt-cotton).
- **Trichoderma:** Free-living fungi found in plant roots that serve as effective biological control agents against multiple plant pathogens.

- **Baculoviruses (Genus: *Nucleopolyhedrovirus*):** These are exceptional narrow-spectrum bio-insecticides. They only attack specific insect pests and arthropods and have zero negative side effects on plants, mammals, birds, or non-target insects. They are highly desirable in Integrated Pest Management (IPM) programs.

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6. Microbes as Biofertilizers

Biofertilizers are living organisms that enrich the nutrient quality of the soil and promote sustainable organic farming, eliminating the need for harmful chemical fertilizers.

- **Symbiotic Bacteria: Rhizobium** bacteria live in the root nodules of leguminous plants, fixing atmospheric nitrogen into organic forms that the plant can use as a nutrient.
- **Free-living Bacteria: Azotobacter** and **Azospirillum** live freely in the soil and absorb free nitrogen from the air to enrich soil fertility.
- **Fungi (Mycorrhiza):** Fungi, particularly from the genus **Glomus**, form symbiotic associations with plant roots. The fungi aggressively absorb **phosphorus** from the soil and pass it to the plant. They also give the plant resistance to root-borne pathogens and increased tolerance to drought and salinity.
- **Cyanobacteria:** Autotrophic microbes like **Anabaena, Nostoc,** and **Oscillatoria** fix atmospheric nitrogen. They are extensively used in paddy (rice) fields because they supply essential nitrogen and add organic matter to the soil.

Part 1: Short-Answer Quiz

Instructions: Answer the following ten questions in 2–3 sentences.

1. **How does Lactic Acid Bacteria (LAB) improve the nutritional quality of curd?**
2. **What are "flocs," and what is their function in secondary sewage treatment?**
3. **Differentiate between the primary and secondary stages of sewage treatment.**
4. **Why was Penicillin referred to as a "wonder drug" during World War II?**
5. **Explain the medical importance of Statins and identify their source organism.**
6. **How does *Bacillus thuringiensis* act as a biocontrol agent against caterpillars?**
7. **What is Biochemical Oxygen Demand (BOD), and how does it indicate water pollution levels?**
8. **Describe the role of methanogens in both ruminants and biogas production.**
9. **How do mycorrhizal fungi from the genus *Glomus* benefit their host plants?**
10. **Why are bottled fruit juices from the market clearer than those prepared at home?**

Part 2: Quiz Answer Key

1. **Lactic Acid Bacteria (LAB)** grow in milk and convert it into curd by producing acids that coagulate and partially digest milk proteins. This process improves nutritional quality by significantly increasing the content of Vitamin B12.

2. **Flocs** are mesh-like masses formed by bacteria associated with fungal filaments. During secondary treatment, these microbes consume the major part of the organic matter in the effluent, thereby reducing the Biochemical Oxygen Demand (BOD).
3. **Primary treatment** is a physical process involving the removal of large and small particles through filtration and sedimentation. **Secondary treatment** is a biological process where aerobic and anaerobic microbes decompose organic matter to make the water less polluting.
4. **Penicillin**, discovered by Alexander Fleming and further developed by Ernest Chain and Howard Florey, was used extensively to treat wounded American soldiers during the war. Its unprecedented ability to combat bacterial infections earned it the reputation of a life-saving "wonder drug."
5. **Statins** are produced by the yeast *Monascus purpureus* and serve as blood cholesterol-lowering agents. They function by competitively inhibiting the enzymes responsible for the synthesis of cholesterol in the liver.
6. **Bacillus thuringiensis** produces toxic protein crystals that, when ingested by insect larvae, become active in the alkaline gut and kill the pest. Biotechnology has allowed scientists to insert these Bt genes directly into crops like cotton, making them resistant to bollworm infestations.
7. **Biochemical Oxygen Demand (BOD)** measures the amount of oxygen consumed by bacteria to oxidize all organic matter in one liter of water. A higher BOD indicates a greater concentration of organic waste and, consequently, a higher potential for pollution.
8. **Methanogens** (such as *Methanobacterium*) are anaerobic bacteria found in the rumen of cattle and in anaerobic sludge digesters. They break down cellulosic material to produce biogas, a mixture consisting primarily of methane, carbon dioxide, and hydrogen.
9. Fungi from the genus ***Glomus*** form a symbiotic association called mycorrhiza, which helps plants absorb phosphorus from the soil. Additionally, they provide the host plant with increased resistance to root-borne pathogens and enhanced tolerance to salinity and drought.
10. Marketed juices are clearer because they are treated with **pectinases and proteases**. These enzymes break down the pectins and fibers that typically cause homemade juices to appear cloudy.

Part 3: Essay Format Questions

Instructions: Use the provided source material to draft comprehensive responses for the following topics. (Answers not provided).

1. **Comprehensive Sewage Management:** Discuss the sequential stages of sewage treatment (Primary, Secondary, and Tertiary), highlighting the specific physical, biological, and chemical processes used to ensure safe discharge into natural water bodies.
2. **Microbes in Sustainable Agriculture:** Evaluate the advantages of using biofertilizers (such as *Rhizobium*, *Azotobacter*, and Cyanobacteria) and biocontrol agents over conventional chemical fertilizers and pesticides.

3. **Industrial Microbiology:** Detail the production and medical applications of bioactive molecules and enzymes, specifically addressing streptokinase, cyclosporin A, and various organic acids.
4. **The Biogas Ecosystem:** Describe the structure and functioning of a typical biogas plant, the role of methanogens, and the historical contribution of the IARI and KVIC in developing this technology in India.
5. **Holistic Pest Management:** Explain the concept of Integrated Pest Management (IPM) and how the use of species-specific agents like baculoviruses supports ecologically sensitive areas and organic farming.

Part 4: Glossary of Key Terms

Term,Definition
Activated Sludge,"The sediment formed in a settling tank during secondary treatment, consisting of bacterial flocs."
Anaerobic Sludge Digesters,Large tanks where anaerobic bacteria digest the organic mass and aerobic microbes of the activated sludge to produce biogas.
Antibiotics,Chemical substances produced by some microbes that can kill or retard the growth of other disease-causing microbes.
Baculoviruses,"Pathogens (specifically of the genus Nucleopolyhedrovirus) used as biocontrol agents for their species-specific, narrow-spectrum insecticidal applications."
Bioactive Molecules,"Microbe-derived molecules that modify metabolism and offer medical benefits, such as cyclosporin A and statins."
Biofertilizers,"Organisms (bacteria, fungi, and cyanobacteria) that enrich the nutrient quality of the soil and enhance plant growth."
Biogas (Gobar Gas),"A mixture of gases, primarily methane, produced by the microbial degradation of organic matter (like cattle dung) in anaerobic conditions."
BOD (Biochemical Oxygen Demand),The amount of dissolved oxygen required for microorganisms to break down the biodegradable organic matter in a water sample.
Clot Buster,"An enzyme, such as streptokinase, used to remove blood clots from the vessels of patients who have suffered myocardial infarction."
Cyanobacteria,"Autotrophic microbes (e.g., Nostoc , Anabaena) that fix atmospheric nitrogen, particularly useful as biofertilizers in paddy fields."
Fermentors,Large industrial vessels used for growing microbes at a scale necessary to produce products like beverages and antibiotics.
Immunosuppressive Agent,A substance like Cyclosporin A that suppresses the immune system to prevent organ rejection in transplant patients.
IPM (Integrated Pest Management),A pest control strategy that uses biological methods and natural predation to keep pests at manageable levels while protecting the ecosystem.
Mycorrhiza,A symbiotic association between fungi (often of the genus Glomus) and plant roots that aids in nutrient absorption and provides pathogen resistance.
Primary Effluent,The supernatant liquid that remains after the physical sedimentation of grit and floating debris during primary sewage treatment.